



1. A *name* for a polygon is a sequence of the polygon's vertices in order. For example, if *AIME* is a rectangle, then *MIAE* is also a valid name for that rectangle, but *MEIA* is not. Let there be N distinct names for the regular 2025-gon $A_1A_2 \dots A_{2025}$ (including $A_1A_2 \dots A_{2025}$ itself). Find the remainder when N is divided by 1000.
2. Let P and Q be the intersection points of the parabolas $y = -x^2 + 21x - 20$ and $y = x^2 - 20x + 21$. The line through P and Q intersects the y -axis at the point $(0, \frac{m}{n})$ for relatively prime positive integers m and n . Find $m + n$.
3. Find the sum of all integers $n \geq 3$ such that the remainder when 2021_3 is divided by n in base 10 is equal to the remainder when 2021_n is divided by 3 in base 10. (Here, 2021_k refers to the number with base- k representation 2021.)
4. For a positive integer N , let $S(N)$ denote the sum of the digits of N and let $P(N)$ the product of the digits of N . Find the number of three-digit positive integers N for which $S(N) + P(N)$ is divisible by 3.
5. Let $ABCD$ be a square and let points P and Q be on segments AB and BC , respectively. Let R be the foot of the altitude from D to segment PQ . Suppose that $PR = 1$, $QR = 2$, $DR = 3$, and the area of square $ABCD$ is $\frac{m}{n}$ for relatively prime positive integers m and n . Find $m + n$.
6. A rat on the coordinate plane starts from the point $(0, 0)$. Every second, if the rat is currently on the (x, y) , it moves to one of the points $(x + 1, y)$, $(x, y + 1)$, $(x + 1, y - 1)$, $(x - 1, y + 1)$.

The rat may not visit the same point more than once, and every point (x, y) the rat visits satisfies $0 \leq x, y \leq 4$. Let N be the number of ways for the rat to move from $(0, 0)$ to $(4, 4)$. Find the number of positive integer divisors of N .
7. Find the number of positive integers n such that both n and $5n$ are five-digit integers with only odd digits.

8. Points P and Q lie on sides AB and CD , respectively, of square $ABCD$ such that $AP = CQ = 1$ and $BP = DQ = 7$. The circumcircle of $\triangle BPQ$ intersects $\overline{BC}$ again at X , and the circumcircle of $\triangle DPQ$ intersects $\overline{DA}$ again at Y . Then, $XY = \frac{a\sqrt{b}}{c}$, where a and c are relatively prime positive integers and b is a positive integer not divisible by the square of any prime. Find $a + b + c$.

9. Let N be the number of permutations $(a_1, b_1, a_2, b_2, \dots, a_6, b_6)$ of $(1, 2, \dots, 12)$ with the following property: it is possible to choose one number among a_i and b_i for each $1 \leq i \leq 6$ such that the six chosen numbers form an increasing sequence and the six remaining numbers form a decreasing sequence. For example, $(1, 11, 10, 3, 7, 6, 5, 8, 9, 4, 12, 2)$ is one such permutation since we can choose $(1, 3, 6, 8, 9, 12)$ and leave $(11, 10, 7, 5, 4, 2)$. Find the remainder when N is divided by 1000.

10. Given that x, y, z are positive integers satisfying

$$(x + y + z)^3 = x^3 + y^3 + z^3 + 24x + 24y + 24z + 48,$$

find the sum of all distinct possible values of $x + y + z$.

11. There are infinitely many boxes in a line, labeled with the integers

$$\dots, -3, -2, -1, 0, 1, 2, 3, \dots$$

in that order. One ball is placed in each box whose label has absolute value at most 2021. Every second, one of the 4043 balls is selected at random, and a coin is flipped. If the coin lands heads, the selected ball moves one box to the right, and if it lands tails it moves one box to the left. After 2021 seconds, the expected number of balls in the box labeled '2' is $1 - \frac{m}{n}$, where m and n are relatively prime positive integers. Find the remainder when m is divided by 1000.

12. Let ABC be a triangle with $AB = 5$, $BC = 6$, $CA = 7$. Let O be a circumcenter of $\triangle ABC$ and let P be a point such that $\overline{AB} \perp \overline{BP}$ and $\overline{AC} \perp \overline{AP}$. If lines OP and BC intersect at T , then the length BT can be expressed in the form $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

13. Let α be a positive real number. Define the infinite sequence of positive real numbers $a_0, a_1, a_2, a_3, \dots$ by

$$\alpha^n = \sum_{k=0}^{\lfloor \frac{n}{2} \rfloor} a_{n-2k} \binom{n}{k}$$

for all nonnegative integers n . Given that $a_8 = 47$, find $\lfloor 100\alpha \rfloor$.

14. Among all pairs of positive integers (x, y) such that $x + y$ divides 2025, randomly and uniformly select one of them. The expected value of $\gcd(x, y)$ is $\frac{m}{n}$, where m and n are relatively prime positive integers. Find the remainder when $m + n$ is divided by 1000.

15. Let ABC be an acute triangle with $AB = 2$ and $AC = 3$. Let O be its circumcenter and let M be the midpoint of $\overline{BC}$. It is given that there exists a point P on the circumcircle of $\triangle BOC$ such that $\angle APB = \angle APC$ and $\angle AOM = \angle APM$. Then $BC^2 = \frac{m}{n}$ for relatively prime positive integers m and n . Find $m + n$.

Time limit: 180 minutes